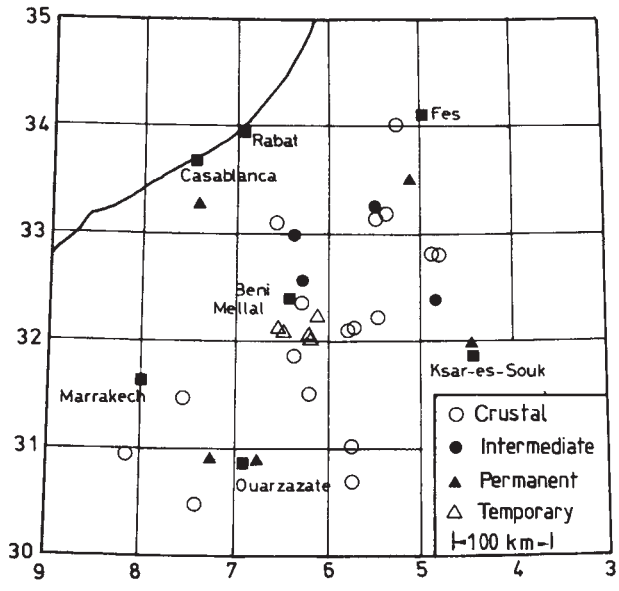


Fig. 4 Epicentres of a 15-day microearthquake survey in the High Atlas. Only events with a difference in arrival of P and S >5 s and error in depth <25 km are shown. ○●, Earthquakes; △▲, seismological stations.

To check the intermediate seismicity beneath the High Atlas, we carried out a microearthquakes survey for 15 days in 1978[13]. The network of six stations had an aperture of 30 km. During this period, 99 events were recorded in at least four stations; 43 of them have a travel time difference between P and S of <5 s and are shallower than 20 km. Among the 56 others, 51 are shallower than 20 km, one is 27 km deep and the last four are deeper than 110 km (Fig. 4). Their magnitude ranges between 1.7 and 2.5 and the closest station is at a mean distance of 60 km. The tests described above applied to these four events show a variation of 25 km in depth and 30 km in epicentre.

Two of the three zones where there are reliable intermediate earthquakes are close to the oceanic boundary between Africa and Europa and could be related to subduction (Gulf of Cadiz and Alboran Sea). On the other hand, the High Atlas whose basement and surroundings are continental is considered a typical intracontinental chain[14]. The existence of basaltic intrusions of Jurassic age[15] is probably related to an extension episode into an entirely continental surrounding[14,16].

Other zones, with intermediate depth earthquakes, are known in the Alpine–Himalayan belt such as Romania, Hindu-Kush, Tibet and Burma. In the Hindu-Kush and Romania a gap in seismicity is observed between 20 and 70 km and the activity decreases at 150 km and disappears at 200 km. This is similar to the results we observe in the High Atlas and differs from that in our two oceanic areas. Although we cannot exclude the possibility that deep seismicity represents the remains of subducting oceanic crust in all of these regions, it is open to question. In the High Atlas particularly, the possibility of a subduction zone in the past 20 Myr seems impossible.

We thank D. Bensari and H. Msougar for help in Morocco, J. Bonnin and M. Mattauer for valuable comments; and especially P. Molnar for his encouragements and suggestions. This research was supported by INAG, contracts ATP 32-01 and 32-23.

Received 30 March; accepted 2 June 1981.

1. McKenzie, D. P. *Geophys. J. R. astr. Soc.* **30**, 109 (1972).
2. Udias, A. & Lopez-Arroyo, A. *Tectonophysics* **31**, 259 (1976).
3. Udias, A. *Rock Mechan.* Suppl. **9**, 75 (1980).
4. Hatzfeld, D. *Ann. Géophys.* **32**, 71 (1976).
5. Hatzfeld, D. thesis, Univ. Grenoble (1978).
6. Munuera, J. M. *El Mapa de Zonas Sismicas Generalizadas de la Peninsula Iberica* (Instituto Geografico y Catastral, Madrid, 1969).
7. Hatzfeld, D. & Bensari, D. *Bull. Soc. geol. Fr.* **7**, 749 (1977).
8. Hatzfeld, D. *C. r. hebd. Séanc. Acad. Sci., Paris* **283**, 1021 (1976).
9. Chatelain, J. L., Roeckers, S. W., Hatzfeld, D. & Molnar, P. *J. geophys. Res.* **85**, 1365 (1980).
10. Banda, E. & Ansorge, J. *Geophys. J. R. astr. Soc.* **63**, 515 (1980).
11. Goula, X., DEA, Univ. Grenoble (1976).
12. Udias, A. & Lopez-Arroyo, A. *Tectonophysics* **9**, 323 (1970).
13. Frogneaux, M. thesis, Univ. Grenoble (1980).
14. Mattauer, M., Tapponnier, P. & Proust, F. *Bull. Soc. geol. Fr.* **7**, 521 (1977).
15. Hailwood, E. A. & Mitchell, J. G. *Geophys. J. astr. Soc.* **24**, 351 (1971).
16. Michard, A. *Notes et Mémoires du Service Geologique, Rabat*, 252 (1976).

Palaeomagnetism of the Ibero-Armorican arc and the Hercynian orogeny in Western Europe

H. Perroud & N. Bonhommet

Laboratoire de Geophysique Interne—Centre Armoricain d'Etude Structurale des Socles, (CNRS), Universite de Rennes I, Campus de Beaulieu, 35042 Rennes Cédex, France

Van der Voo[1] has pointed out that further advances in the understanding of the Hercynian orogeny require palaeomagnetic data from well studied orogenic areas. We report here palaeomagnetic results of Palaeozoic rocks taken in Brittany, Portugal and Spain in the Ibero-Armorican arc (IAA), a large geological feature considered to be a key area for the Hercynian orogeny in Western Europe. They show evidence that the IAA formed in two stages, extending results obtained in Spain by Ries *et al.*[2]. Giving precise constraints for the former shape of the arc we can use pre-Hercynian palaeomagnetic directions found in Spain and Portugal to follow its evolution during Palaeozoic times. This megastructure linking Brittany to Iberia, squeezed between Gondwana and the northern continents during the Carboniferous, did break away from the former block during Ordovician times.

Geological[3,4] and geophysical[5] evidence suggests that the IAA is formed of large juxtaposed curved units lying on the two sides of the Bay of Biscay. Stratigraphic studies of Palaeozoic formations from the arc[6,7], performed at the Rennes CNRS centre, suggest that the shape of this structure could have been quite different before the Hercynian deformation. In particular, the very strong correlations between the stratigraphy and palaeontology of the Crozon area (Brittany) and the Buçaco syncline (Portugal) for Ordovician rocks imply that these two regions belonged to the same sedimentary basin in the Lower Palaeozoic, and thus palaeogeographic reconstructions[7] tend to place these two regions next to each other. Moreover, the question of whether the arc is of primary or secondary origin is crucial for several of the models proposed for the Hercynian orogeny. A primary origin implies that the arc always had the same curvature whereas a secondary origin implies oroclinal bending following Carey's model[8]. Ries *et al.*[2] recently showed that the Iberian arc, the inner part of the IAA, had a predominant secondary origin. The absence of definitive evidence that the two structures did evolve simultaneously led us to extend the sampling to the whole IAA and to different stages in the Palaeozoic.

Samples were from: (1) Brittany: Upper Ordovician dolerites of Crozon (14 sites, 100 samples); (2) Spain: Ordovician volcanics of Cabo de Penas (2 sites, 31 samples), Upper Devonian red sandstones of Candas (3 sites, 32 samples), Lower Carboniferous red grits, Alba formation from Candas (3 sites, 23 samples) and San Emiliano (6 sites, 42 samples); (3) Portugal: Cambro-Ordovician redbeds (7 sites, 60 samples) and Upper Ordovician dolerites (4 sites, 34 samples) from Buçaco (Fig. 1). Natural remanent magnetization (NRM) was measured for these samples and for most of them a viscosity test[9] was also performed; a.c. field, thermal and chemical demagnetizations were carried out (all samples being treated by at least one of these methods, mostly by two). In addition, rock magnetic experiments such as coercivity spectrum analysis[10] of pairs of

0028-0836/81/310445—04$01.00
© 1981 Macmillan Journals Ltd

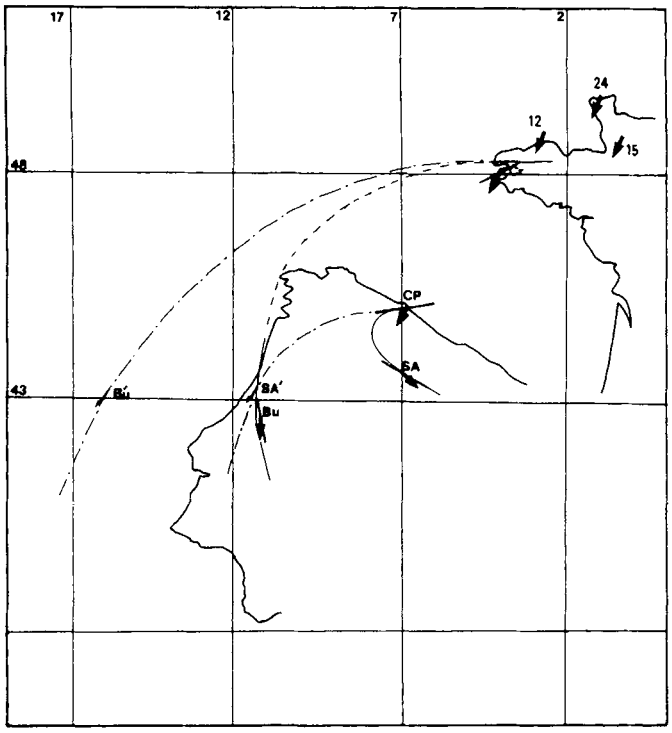


Fig. 1 Palaeomagnetic Carboniferous directions (thick arrows) found on the Ibero-Armorican arc with sampling locations (CR, Crozon; CP, Cabo de Penas and Candas; SA, San Emiliano; BU, Buçaco) after closing of the Bay of Biscay (fitted by Lefort[24]). Values for additional data in Brittany correspond to the references given in the text (see also Table 1). The ante-tectonic shape of the arc deduced from the data is also given with the corrected Carboniferous directions (small arrows).

leached and unleached samples, heating to Curie temperature or cooling to the liquid air temperature of an acquired induced remanent magnetization (IRM), were carried out to support the interpretation of the multivectorial magnetization with physical arguments on the magnetic carriers. The results obtained from these studies[11], which will be published in detail elsewhere, can be summarized as follows.

The magnetic behaviour of the Upper Ordovician dolerites from Buçaco and Crozon is identical. Intensities are low, with a maximum frequency between 10^{-2} and 10^{-3} A m^{-1} and viscosities are high (>20% for 50% of the samples); the IRM acquisition curves are identical; they display the same magnetic transition at −160 °C, suggesting the presence of magnetite, and current work at the Rennes CNRS centre gives K–Ar ages which range for both regions from 190 to 300 Myr. In accordance with the radio-dating results, the remanent magnetization is interpreted as being secondary with an acquisition period between Carboniferous and Triassic or Jurassic time (Table 1), thus confirming the geological relationships between Crozon and Buçaco.

Buçaco redbeds exhibit magnetization directions interpreted as pre-tectonic and remagnetization directions as being from the Upper Cretaceous and Quaternary[11]. A characteristic of the pre-tectonic directions is a constant declination of 150° and variable inclination. The same behaviour was found by Duff[12] in an extensive palaeomagnetic survey of redbeds in Brittany with ages extending from Lower Cambrian to Lower Ordovician. From this work two sites at Crozon are well suited for comparison with Buçaco as both areas are thought to be part of the same sedimentary basin[7], and thus allow comparison with data from contemporaneous redbeds. Components B3 and C found by Duff[12] on 10 specimens of these two sites using acid leaching, thermal and a.f. cleaning, gave a mean direction of 220° after tectonic correction (sites CP, CR in Tables 2 and 3 of ref. 12). The difference in declination between the two areas suggests a partial secondary origin of the IAA if we correct for the rotation of Spain during the Cretaceous and look at the variation of strike of the arc. Again, the palaeomagnetic characteristics of contemporaneous samples from Brittany and Portugal are identical.

At Cabo de Penas, Ordovician volcanics show two well defined pretectonic directions of magnetization (Table 1). One (component A) has a very steep inclination, coercivities over 70 mT and unblocking temperatures over 500 °C, and has been interpreted as Ordovician; the other (component B) has a shallow inclination and is comparable with the direction obtained east of Cabo de Penas for the Upper Devonian and Lower Carboniferous redbeds of Candas (component A); the

Table 1 Palaeomagnetic data from the Ibero-Armorican arc

Site (reference on Fig. 1)	Rock unit	Component of magnetization	N	D	I	k	α	$\bar{S}$	Age of magnetization	Ref.
South limb										
SA	San Emiliano redgrits	—	30	102	13	53	2	270°	Lower Carboniferous	14
BU	Buçaco dolerites	—	20	139	−43	58	4	145	Jurassic ?	This work
	Buçaco redbeds	A	7	151	80	83	6	145	Ordovician	This work
		B	7	143	61	41	9	145	Between Ordovician and	
		C	6	144	36	45	9	145	Lower Carboniferous ?	This work
		D	3	153	5	212	6	145	Lower Carboniferous	
North limb										
CP	Cabo de Penas volcanics	A	13	208	78	50	6	230	Ordovician	This work
		B	78	178	19	39	8	230	Lower Carboniferous	This work
	Candas redbeds	A								
		B	8	185	23	27	10	230	Middle–Upper Carboniferous	This work
CR	Crozon dolerites	A	19	208	−15	215	2	240	Triassic	This work
		B	38	205	29	33	4	240	Middle Carboniferous	This work
		C	12	350	−11	300	3	240	Upper Carboniferous	This work
	Armorican redbeds	B3	10	216	41	35	10		Devonian to early Carboniferous	12
		C	16	228	63	70	7		Silurian to early Carboniferous	12
15	Montmartin redbeds	—	186	208	16	20	7	270	Late Devonian–early Carboniferous	15
24	Flamanville granite	—	8	203	13		14		Carboniferous	22
12	Ploumanach granite (300 Myr)	—	31	200	9	121	7		Middle Carboniferous	12
	Jersey dolerites	A	—	199	16	71	9		Middle–late Carboniferous	16
		B	—	204	57	59	8		Siluro-Devonian	16
	Jersey volcanics (533 Myr)		66	292	78	8	19		Cambrian	16

Palaeomagnetic component directions found on the Ibero-Armorican arc. N, Number of samples used; D, I, declination, inclination; k, α_{95} Fisher statistical parameters[23]; $\bar{S}$, mean structural direction. Statistics may refer to samples, specimen, directions or sites: see in each case original work given in the reference; only the total number of samples used is given here; all data were cleaned using a.f., acid and thermal cleaning as done by Duff[12,16], N.B. *et al.*[14] and Jones *et al.*[15] and a.c. field up to 0.2 T (2,000 oe) as in ref. 22.

©1981 Nature Publishing Group

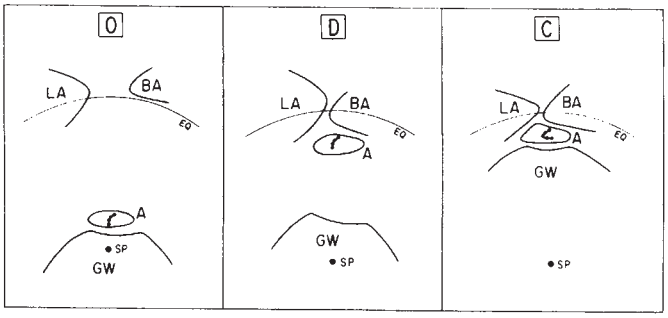


Fig. 2 Scheme of the evolution of the main plates involved in the Hercynian orogeny in Palaeozoic times (O, Ordovician; D, Devonian; C, Carboniferous) with mention of the Ibero-Armorican arc. LA, Laurentia; BA, Baltica; GW, Gondwana; A, Armorica.

eight sites (volcanics and sedimentary) were used to give a mean direction which is pre-tectonic as shown by a fold test. We can do better than the classical tilt correction on this example to calculate the mean direction: if we look at the angle between the strike direction and the magnetization on each site, this angle seems to be very constant (52.5°± 5°) regardless of the site and even though strike directions vary on this example between 210° and 244°. This suggests that deformation occurred in two stages, the latter one with a vertical axis correlated with the tectonic closure of the arc[13], and that the magnetization was acquired before deformation; taking this result into account and using a regional mean direction for strike of 230°, the mean palaeomagnetic direction of the eight sites in Table 1 has been obtained after correcting each site for strike variation. The six sites of redbeds near Candas also show a soft component obtained on a few samples but scattered among all the sites (Candas redbeds component B, Table 1); the post-tectonic origin of this component is clearly indicated by a negative fold test[11]. The corresponding inclination is very close to that obtained at Crozon on the dolerites (Crozon dolerites component B, Table 1); this last component has also been interpreted as post-tectonic but with a magnetization predating the closure of the arc, as shown by using a 'strike direction' correction on the 14 sites of Crozon. Magnetization was therefore acquired at Crozon between two main phases of deformation of the IAA.

The direction given in Table 1 for San Emiliano was obtained by N.B. *et al.*[14] using an unfolding model which took into account the two phases of deformation in the area; the effect of the model was to increase the precision parameter k (from 14 to 64) in comparison with the usual tilt correction. A pre-tectonic direction was found (Table 1) and a Lower Carboniferous age for the magnetization inferred from the age of the sediments.

Ries *et al.*[2] sampled the same Alba formation at Carranques near Candas and at San Emiliano; however, their result cannot be used for comparison because only three samples at Carranques were thermally cleaned and the authors report a direction $D = 211°$, $I = 48°$ with $k = 2$ showing large scatter after thermal cleaning; this is a general behaviour of all results obtained after cleaning as reported in that study. At San Emiliano four sites have been studied for the NRM but none appears among the thermal cleaned results given in Table 2 of their study.

Carboniferous data from Brittany obtained by Jones *et al.*[15] on the syncline of Montmartin and by Duff[12] from the Ploumanach granite, Jersey dolerites and volcanics, together with our own results at Crozon and Candas (CR, CP, Fig. 1) reveal (Table 1) that the corresponding directions are not significantly different for the northern part of the IAA before, (Candas A) and after (Candas B, Crozon B) deformation: most of the rotation seems to have occurred in the southern part of the arc. A new shape of the IAA can then be inferred taking the simplest compatible solution with the palaeomagnetic data. This indicates that 80° of the curvature of the arc are secondary and that 70° are primary or pre-Carboniferous.

Ries *et al.* used results based on NRM directions from 29 sites along the Iberian arc to demonstrate convincingly the partially secondary origin of this structure; however, because these authors did not consistently use cleaned results to look for multicomponent magnetization, because they infer a Variscan age for all magnetizations, their data cannot be used for comparison as they give no directions associated with the age of magnetization for the different rock units studied. However, their result, a bending of 110°, is not very different from the 80° reported here.

From the new position of the sampling areas and the palaeomagnetic vectors calculated for the new shape of the arc (that is, after elimination of the Hercynian deformation), we can determine virtual palaeomagnetic poles for our Ordovician sites. The Buçaco (25°N, 335°E) and Cabo de Penas (30°N, 330°E) poles coincide in the new reconstruction, indicating that the shape of the arc we deduced from the Carboniferous data is also compatible with our Ordovician results. We therefore propose that 70° of the arc are effectively primary (presedimentary), inherited from Precambrian times. The mean Carboniferous declination of the north limb of the IAA differs by a few degrees (10°–15°) from that of stable Europe as shown by Jones *et al.*[15] for Montmartin syncline; it is not clear, however, whether the bending took place entirely in the southern part of the arc with a further rotation of 15° of the whole Ibero-Armorican block or whether it was the closure of the arc that caused the 15° difference with Europe. However, this does not change the former shape of the IAA found from the data. The Crozon B and Candas B components, both post-tectonic but ante-closure of the arc, in agreement with results of Jones *et al.* at Montmartin, show that at that time the IAA with its primary shape was close to Europe, with a large ocean separating it from Gondwana, a result shown for Brittany by Jones *et al.*[15] (Fig. 2).

The Ordovician mean palaeomagnetic pole obtained (27°N, 332°E) falls between Cambrian and Siluro-Devonian poles found by Duff[16] for the Armorican massif and this agrees with the Ordovician age assumed for the magnetization. Interestingly, this pole is quite different from conventional Ordovician poles for the British Isles[17] but could more easily be correlated with north Wales data, which have until now been considered as anomalous[18]. Recent results by Hagstrum *et al.*[19] showing that the apparent polar wander curve of the Armorican massif is very similar to the Eo-Cambrian–Cambrian curve for Gondwanaland prompted us to compare the Ordovician palaeolatitude for the IAA (70°) with the African Ordovician pole[20]; this shows that the IAA should have been very near to Gondwanaland in the early Palaeozoic. These results show, moreover, that the Armorican massif and Iberia Meseta were probably a single unit during the entire Palaeozoic, thus justifying the name of 'Armorica plate' given by Van der Voo[21] to these regions of Western Europe; the large drift of this plate since the Ordovician shows that collision and subduction tectonic processes did exist during the Hercynian orogeny. Finally, taking into account all these results and using available data given by Van der Voo[1], we propose a scheme of the main plates involved in the Hercynian orogeny with particular reference to the Armorica plate bearing the IAA (Fig. 2).

The research was supported by a grant from INAG through 'ATP Géodynamique' national programme. We thank Dr Van der Voo for suggestions and comments.

Received 2 February; accepted 2 June 1981.

1. Van der Voo, R. *et al. Int. geol. Congr. Colloq.* C6, 204–212 (1980).
2. Ries, A. C. *et al. Earth planet. Sci. Lett.* **50**, 301–310 (1980).
3. Cogné, J. *Histoire Structurale du Golfe de Gascogne* (IFP) I1,1, I1–23 (1971).
4. Matte, P. *Nova Acta Leopoldina* **224**, 239–262 (1974).
5. Lefort, J. P. *Geology* **7**, 384–388 (1979).
6. Henry, J. L. *et al. Commun. Serv. Géol. Port* **57**, 303–345 (1974).
7. Paris, F. & Robardet, M. *Bull. Soc. Géol. Fr.* **7**, 1121–1126 (1977).
8. Carey S. W. *The Tectonic Approach to Continental Drift* (ed. Carey, S. A.) (University of Tasmania, 1958).
9. Thellier, E. & Thellier, O. *Ann. Geophys.* **15**, 285–376 (1959).
10. Dunlop, D. J. *Geophys. J. R. astr. Soc.* **27**, 37–56 (1972).

©1981 Nature Publishing Group

11. Perroud, H. thesis, Univ. Rennes (1980).
12. Duff, B. A. *Geophys. J. R. astr. Soc.* **59**, 345–365 (1979).
13. Julivert, M. *Histoire Structurale du Golfe de Gascogne (IFP)* I2-1, I2-28 (1971).
14. Bonhommet N. *et al. J. geophys. Res.* **86**, 1873–1887 (1981).
15. Jones, M. *et al. Geophys. J. R. astr. Soc.* **58**, 287–308 (1979).
16. Duff, B. A. *Geophys. J. R. astr. Soc.* **60**, 355–375 (1980).
17. Piper, J. *Earth planet. Sci. Lett.* **44**, 176–192 (1979).
18. Thomas, C. & Briden, J. C. *Nature* **259**, 380–382 (1976).
19. Hagstrum, J. T. *et al. Geophys. J. R. astr. Soc.* **61**, 489–517 (1980).
20. Morel, P. & Irving, E. *J. Geol.* **86**, 535–561 (1978).
21. Van der Voo, R. *EOS Trans.* **60**, 241 (1979).
22. Van der Voo, R. & Klootwijk, C. T. *Geologie Mijn.* **51**, 609–617 (1972).
23. Fisher, R. A. *Proc. R. Soc.* **A217**, 295–305 (1953).
24. Lefort, J. P. *Mar. Geol.* **37**, 355–369 (1980).

Dietary carbon sources of mussels and tubeworms from Galapagos hydrothermal vents determined from tissue ^{14}C activity

P. M. Williams*, K. L. Smith*, E. M. Druffel† & T. W. Linick†

* Scripps Institution of Oceanography and
† Department of Chemistry, University of California, San Diego, La Jolla, California 92093, USA

The large quantities of reduced carbon that are required to support the filter-feeding mytilid mussels (*Mytilus* sp.), vesicomyid clams (*Calyptogena* sp.) and various other animals in the Galápagos hydrothermal vent systems are thought to be derived from either the *in situ* synthesis of particulate organic matter by chemoautotrophic, sulphide-oxidizing bacteria[1,2] or by the advection of sedimentary organic carbon into the vent environment from surrounding areas[3,4]. In contrast, the dense populations of vestimentiferan tubeworms (*Riftia pachyptila*), which lack mouth organs and digestive tracts, apparently utilize organic carbon synthesized by symbiotic chemoautotrophs[5]. We present evidence here, based on ^{14}C activities and $^{13}C/^{12}C$ ratios, that the principal source of dietary carbon for mussels and tubeworms is derived from the dissolved inorganic carbon (DIOC) in the vent effluent waters.

There are two postulated sources of particulate organic carbon (POC) available as dietary carbon to filter-feeding organisms. The first is sedimentary POC derived from DIOC fixed photosynthetically at the ocean's surface which subsequently reaches the ocean floor. Its radiocarbon activity is assumed to be the same as that of the DIOC in surface water ($+20 \pm 20$, Table 1). The $\delta^{13}C$ of this POC (-22, Table 1) is a mean value derived from $\delta^{13}C$ measurements of POC[6,7] collected below 500 m. The second source is POC synthesized chemoautotrophically from DIOC in the vent waters. This DIOC originates from both magmatic activity[8] ($\Delta^{14}C = -1{,}000$ or 'dead' carbon) and the ambient bottom water ($\Delta^{14}C = -233$, Table 1) which mixes with the high-temperature hydrothermal fluid. As the organic ^{14}C activities of the organisms grown in the vent systems reflect the $\Delta^{14}C$ of these three carbon sources and the $^{13}C/^{12}C$ ratios reflect the carbon isotope fractionation occurring during synthesis of organic tissues in these organisms, it is possible to calculate the relative contribution of each carbon source using the following equations:

$$\Delta^{14}C_{mussel} = x(-1{,}000) + y(-233) + z(20) \quad (1)$$

$$\delta^{13}C_{mussel} = (x + y)(\delta^{13}C_{vent\ POC}) + z(-22) \quad (2)$$

where x, y and z are the relative amounts of magmatic DIOC, ambient DIOC and sedimentary POC, respectively. The $\delta^{13}C$ of POC in the vent water is unknown. Both of the above equations have to be satisfied to determine the relative amounts of the three possible dietary carbon sources utilized by the filter-feeding organisms.

The quantitative contribution of sedimenting POC to the food supply for filter-feeding vent organisms is questionable considering the flux of POC to the ocean floor and the *in situ* respiration rates of mussels. The total, mean flux of POC to the sea floor at 2,670 m 0°35.75′ N, 86°05.66′ W measured 20 and 100 m above the bottom on 4 July 1976 was 2.0 g C $m^{-2}yr^{-1}$ (ref. 9). *In situ* measurements of individual mussel respiration rates by K. Smith (unpublished results) range from 0.30 to 0.81 ml O_2 h^{-1} in dense mussel beds. Assuming a respiratory quotient of 0.85, then these rates are equivalent to an organic carbon requirement of 1.2–3.3 g C yr^{-1} (mean = 2.3 g C yr^{-1}). Thus, each mussel would essentially require the total annual POC flux per square metre for maintenance with no growth.

The ^{14}C activity of the mussel tissue could result from a mixture of sedimentary POC and POC of low ^{14}C activity derived from DIOC in the vent waters. The maximum amount of magmatic DIOC in the Galápagos vent waters ($T = 17$ °C) has been estimated to be 375 μmol kg^{-1}, a 13% increase above the ambient DIOC concentration of ~2,400 μmol kg^{-1} (refs 10, 11). If 13% of the vent water DIOC is magmatic ($\Delta^{14}C = -1{,}000$) and the remaining 87% is composed of ambient DIOC ($\Delta^{14}C = -233$) plus sedimentary POC derived from the DIOC in surface water ($\Delta^{14}C = +20$), then the maximum amount of sedimentary POC that could be incorporated into mussel tissue ($\Delta^{14}C = -267$) is 26% (equation 1). Estimates of mussel density in the 'Musselbed' vent area (W. Smithy, personal communication) are 20 ± 9 specimens per 0.25 m^2 in the immediate vicinity of the vent plume. If the mussels are utilizing the calculated maximum of 26% of sedimentary POC for growth (taken here as equivalent to maintenance), then each vent musselbed would use ~48 g C $m^{-2}yr^{-1}$ (0.26×2.3 g C yr^{-1} per mussel $\times$ 80 mussels per m^2), or about 35% of the annual primary productivity per square metre in the euphotic zone[9] for the growth of mussels alone.

There is a significant difference between the $^{13}C/^{12}C$ ratios, as well as ^{14}C activities, of mussel tissue collected 1 and 8 m from the vent plume (Table 1). If sedimentary POC ($\delta^{13}C = -22$) was incorporated into the tissue of the 8-m mussel, the resulting $\delta^{13}C$ value (-32.3) would be less than that observed (-32.8). This calculation assumes that the ^{14}C activity of the 8-m mussel tissue ($\Delta^{14}C = -228$) is derived from a mixture of 14% sedimentary POC ($\Delta^{14}C = +20$) and 86% chemosynthetic POC ($\Delta^{14}C = -267$) (equation 1).

If the ^{14}C activity of mussel shell carbonate reflects the ^{14}C activity of DIOC in the surrounding seawater, then the 21‰ decrease in $\Delta^{14}C$ between the ambient seawater DIOC and the mussel shell carbonate reflects dilution of the ambient DIOC with 3% magmatic DIOC. For the two clam shell carbonates, the 30‰ decrease in $\Delta^{14}C$ would represent dilution of the ambient DIOC with 4% magmatic DIOC, taking the ^{14}C activity of the bottom water at 21° N to be the same as at the Galapagos rift zone. A similar calculation gives 4.4% magmatic DIOC incorporated into cellular carbon synthesized by the chemoautotrophic bacteria and used by the mussels 1 m from the plume (provided the contribution from sedimentary POC is negligible).

The identical ^{14}C activities of the vestimentiferan tubeworm tissue and the mussel tissues collected 1 m from the vents strongly suggest that the tubeworms and mussels are utilizing the same DIOC sources, even though $\delta^{13}C$ in the tubeworm tissue is 23‰ greater than in the mussel tissue (Table 1). These atypical $^{13}C/^{12}C$ ratios found in the vent organisms have been discussed by Rau[12,13]. If bacterial chemosynthesis is the major process providing dietary carbon for the mussels and tubeworms, it is still unclear why carbon isotope fractionation should be different between chemoautotrophic synthesis occurring in the vent waters and internally (symbiotically) in the tubeworms.

An additional carbon source which may contribute to the bacterial synthesis of POC in the vent waters is methane, energetically a more favourable substrate for the growth of heterotrophic bacteria than DIOC. Methane concentrations up to 3.4 μmol kg^{-1} have been measured in the Galápagos vent

0028-0836/81/310448—02$01.00

© 1981 Macmillan Journals Ltd